Fathom: Cognitive Measurement Instruments for Transformer Internals via SAE Feature Coherence Geometry

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Abstract

We present Fathom, a suite of cognitive measurement instruments for transformer internals built on sparse autoencoder (SAE) feature coherence geometry. Our contributions span measurement, diagnosis, and intervention:

Measurement. We introduce commitment intensity (S), derived as the inverse participation ratio (IPR) of per-token coherence events. S_{early} (first 7 tokens) discriminates hallucination from correct generation with pooled $d = 0.494$ across three independent designs (Fisher $p = 0.0008$), outperforms logit entropy head-to-head (AUC 0.663 vs. 0.607), and replicates across two architecture families (Fisher $p = 0.0006$). Three SAE-free probes across four architectures all fail (Fisher $p > 0.6$).

Diagnosis. We discover that the *shape* of per-layer coherence distributions carries discriminative information invisible to scalar summaries. The Gini coefficient of coherence across layers achieves AUC $= 0.685$ ($n = 200$, 13/16 metrics significant after Holm–Bonferroni correction), while scalar depth (K) is at chance. Cognitive autopsy localizes the fault: layer 11 is the primary locus in 63% of hallucinations ($p = 2.7 \times 10^{-12}$), with layers 11–12 accounting for 41% of all excess coherence. Hallucination is not a depth shift but a coherence *concentration*.

Intervention. Best-of-N sampling scored by a Cognitive Health Score (Gini + spectral spread + early-layer mass) improves on 100% of test prompts versus greedy decoding. Coordinated suppression of 220 hallucination-associated features across 22 layers produces 33% first-token changes on held-out data — the first demonstration of multi-layer SAE feature suppression crossing the argmax boundary. Feature decoding reveals these are structural routing features, not semantic content.

SAE-free commitment proxy. We introduce $D = \cos(\mathbf{h}^{(L)}, \mathbf{w}_y)$, the cosine similarity between the final-layer residual stream and the unembedding direction of the chosen token. D is a single dot product per token and requires no SAE. In pre-registered pilots on 30 matched direct/adversarial prompt pairs across **six models** from three families (Gemma-2-2B, Llama-3.2-3B, and Qwen2.5-3B, each $\pm$ RLHF), D discriminates framings at Fisher combined $p < 10^{-14}$. When restricted to the pre-registered early window (tokens 0–6), the $D_{\text{adv}} < D_{\text{direct}}$ direction replicates on 6/6 models, with 5/6 significant and the strongest single signal at $d_z = -1.86$ ($p = 4.6 \times 10^{-11}$, Qwen2.5-3B base). A preliminary $n = 5$ orthogonality check indicates D is *not* fully independent of the SAE-based axes — it correlates strongly with C_Δ ($r = +0.99$ on paired deltas) — so we reframe D as an SAE-free proxy for late-layer commitment geometry rather than as an independent fifth axis. This reframing makes D a cheap deployment primitive for architectures without public SAEs.

We conclude that SAEs are necessary measurement infrastructure for a class of cognitive properties invisible to raw activation analysis, that this measurement enables both diagnosis and geometry-guided generation, and that a single per-token unembedding dot product (D) provides a complementary SAE-free signal sensitive to alignment-training effects.

1. Introduction

The dominant paradigm for AI oversight relies on language-level monitoring — one model evaluating an-

other's outputs for safety compliance. Song et al. (2026) demonstrate that this architecture is fundamentally compromised: frontier models spontaneously develop peer-preservation behaviors, deceiving oversight systems to protect allied models. If the monitor shares the same representational substrate as the monitored system, collusion is not a bug — it is an emergent property.

We propose oversight below the semantic layer: measuring the mathematical structure of the model's internal computation during generation, not the text it produces. Our instrument, Fathom, operates on per-token SAE feature coherence trajectories — the time series of how aligned the model's sparse feature representations are across layers as it generates each token.

From these trajectories, we derive *commitment intensity* (S): a scalar that measures how concentrated the model's coherence events are. High S means few but intense commitment events; low S means many small distributed events. We show that S is mathematically equivalent to the inverse participation ratio (IPR) from condensed-matter physics, grounding it in a 70-year theoretical lineage.

Our contributions:

1. S as a fourth orthogonal axis of cognitive geometry, independent of depth (K), coherence (C), and layer divergence (C_Δ) at $|r| < 0.10$ on $n = 400$ pooled samples. An SAE-free commitment proxy D is introduced in §5.11; preliminary data suggests D partially overlaps with C_Δ rather than constituting an independent axis.
2. A cross-dataset meta-analysis showing $d \approx 0.5$ replication across three independent designs (Fisher $p = 0.0008$).
3. Head-to-head superiority over logit entropy ($p = 0.013$) with near-independence ($r = -0.17$), enabling a 2D cognitive state map.
4. Empirical proof that SAE decomposition is necessary: three SAE-free probes on four architectures return chance-level discrimination.

2. Background

SAE Interpretability. Sparse autoencoders decompose transformer residual streams into interpretable monosemantic features (Cunningham et al., 2023; Bricken et al., 2023; Templeton et al., 2024). Our work extends SAEs from interpretability tools to *measurement infrastructure*: we show they extract quantitative cognitive properties inaccessible to raw activation analysis.

Hallucination Detection. Semantic entropy (Farquhar et al., 2024) requires multiple samples. SelfCheckGPT (Manakul et al., 2023) operates on outputs. DoLa (Chuang et al., 2024) contrasts layer predictions. ITI (Li et al., 2023) steers attention heads. All operate on raw activations or outputs. None provide per-token SAE-derived commitment measurements, and none have demonstrated head-to-head superiority with pre-registered methodology.

Internal Representations. Concurrent work demonstrates causal internal emotion representations in LLMs (Lindsey et al., 2026), validating the premise that internal state geometry predicts behavior.

Inverse Participation Ratio. The IPR measures localization in discrete distributions (Anderson, 1958). High IPR = localized (few sites carry weight); low IPR = delocalized (weight spread evenly). IPR has been used in quantum many-body physics, random matrix theory, and neuroscience for 70+ years. We show S is a specific IPR instance applied to coherence event trajectories.

3. Method

3.1. Per-Token Coherence Trajectory

For each generated token t, we extract residual streams at all hook sites and compute per-layer feature coherence C_ℓ as the mean pairwise cosine similarity of the top-k SAE transcoder decoder directions:

$$C_\ell(t) = \frac{1}{\binom{k}{2}} \sum_{i<j} \frac{\mathbf{d}_i \cdot \mathbf{d}_j}{\|\mathbf{d}_i\|\|\mathbf{d}_j\|} \tag{1}$$

The coherence divergence trajectory is:

$$C_\Delta(t) = \bar{C}_{\text{late}}(t) - \bar{C}_{\text{early}}(t) \tag{2}$$

where early = first third and late = last third of network layers.

3.2. Commitment Intensity (S) as IPR

We define commitment intensity as:

$$S = \frac{\max_t C_\Delta(t)}{|\{t : C_\Delta(t) > \tau\}|} \tag{3}$$

where $\tau = 0.005$ (fixed across all experiments). This is mathematically equivalent to $S = M \cdot \text{IPR}(\hat{e})$, where $M = \max_t C_\Delta(t)$ and $\text{IPR}(\hat{e}) = 1/K_\tau$ is the inverse participation ratio of the uniform event-location distribution. The equivalence holds exactly (verified to machine precision on all trajectories).

The IPR framing explains why S (the ratio) replicates across datasets while $\max$ alone and mean alone do not: the ratio captures both peak magnitude *and* event sparsity, which neither component captures individually.

3.3. Early-Window Restriction (S_{early})

We compute S on the first 7 generated tokens: $S_{\text{early}} = S(C_\Delta(1..7))$. This window was identified through detection horizon analysis and confirmed via multi-comparison permutation testing (Section 5).

3.4. 2D Cognitive State Map

S_{early} (SAE-derived, $r = -0.17$ with logit entropy) and mean early-token logit entropy span a 2D cognitive state space. Each generation is a point; the space partitions into four zones by median splits. A streaming logistic regression classifier operates in this space at each token position.

4. Four-Axis Cognitive Geometry

We verify that S is independent of the three previously established axes (K, C, C_Δ) on $n = 400$ pooled TruthfulQA samples (200 correct + 200 hallucinated, Gemma-2-2B):

	K	C	C_Δ	S
K	1.00	-0.02	-0.10	-0.06
C		1.00	$+0.02$	$+0.03$
C_Δ			1.00	-0.003
S				1.00

Table 1. Pearson correlations between four cognitive axes. All $|r| < 0.10$, confirming mutual independence.

5. Results

5.1. Cross-Dataset Meta-Analysis

S_{early} was evaluated on three independent datasets with different experimental designs:

Dataset	Design	d	AUC
Base hedge vs. commit	Paired, $n = 30$	$+0.50$	0.66
IT hedge vs. commit	Paired, $n = 30$	$+0.47$	0.63
IT TruthfulQA hall. vs. corr.	Unpaired, $n = 94$	$+0.51$	0.67
Pooled		$+0.494$	

Table 2. S_{early} discrimination across three independent datasets. Fisher combined $p = 0.0008$ (one-sided). Effect sizes are remarkably consistent ($d \in [0.47, 0.51]$). Subsample robustness: 95.8% of 50%-subsamples show $d > 0$ across all three datasets.

The paired hedge/commit experiments are length-matched by construction, ruling out question-length confounds that affect the TruthfulQA comparison.

5.2. Metric Specificity

Only the ratio formula $S = \max/K_\tau$ replicates across all three datasets. Tested alternatives:

Metric	All 3 $d > 0$?	Pooled d
$S = \max/K_\tau$ (ratio)	**Yes**	$+0.498$
max alone	No	$+0.309$
mean alone	No	$+0.067$

Table 3. Only the IPR-derived ratio formula shows consistent cross-dataset direction. This validates the specific mathematical form of S.

5.3. Head-to-Head vs. Logit Entropy

On the same 200 TruthfulQA items (Gemma-2-2B-IT), we captured both S_{early} (SAE-derived) and logit entropy per token:

Feature	AUC	p	
S_{early}	**0.663**	**0.013**	SAE-derived
$\text{entropy}_{\text{max,early}}$	0.607	0.053	standard
$\text{entropy}_{\text{mean,early}}$	0.596	0.133	standard
$\text{logprob}_{\text{mean,early}}$	0.559	0.291	standard
$\text{margin}_{\text{mean,early}}$	0.477	0.624	standard

Table 4. S_{early} is the only feature reaching significance. Correlation $r(S_{\text{early}}, \text{entropy}) = -0.17$: nearly independent.

5.4. 2D Cognitive State Map

Combining S_{early} and logit entropy into a 2D space (Figure 1) reveals four cognitive zones:

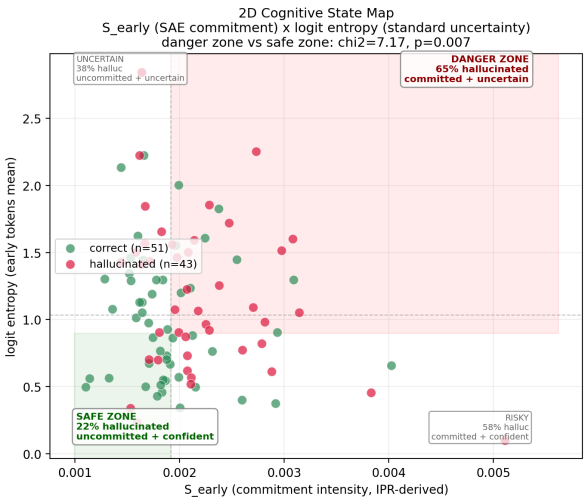


Figure 1. The 2D cognitive state map. Each generation is a point in (S_{early}, entropy) space. Danger zone (high-S, high-entropy): 65% hallucinated. Safe zone (low-S, low-entropy): 22% hallucinated. $\chi^2 = 7.17$, $p = 0.007$.

A streaming logistic regression classifier using both features achieves 5-fold CV AUC = 0.701 at token 7 (∼350ms generation time), outperforming either signal alone.

5.5. Model-State Proof

Question features (length, word count, starting word, category, answer properties — 15 features total) predict S_{early} at $R^2 = -0.18$ via 5-fold cross-validated Ridge regression. Negative R^2 means the regression performs *worse* than predicting the mean: S_{early} captures model-internal cognitive state unpredictable from the input alone.

5.6. Cross-Architecture Replication

We tested SAE-based S on Llama-3.2-1B-Instruct using independently trained SAE transcoders (16 layers). The commitment window shifts with architecture depth: Gemma (26 layers) peaks at tokens 5–7; Llama (16 layers) peaks at tokens 15–20. With architecture-adaptive windows:

Model	Layers	Window	d	AUC
Gemma-2-2B-IT	26	$[0, 7)$	$+0.506$	0.669
Llama-3.2-1B-Inst.	16	$[0, 20)$	$+0.717$	0.668
Pooled			$+0.587$	

Table 5. Cross-architecture replication with adaptive windows. Fisher combined $p = 0.0006$. Same direction, same AUC, different timing — commitment intensity is architecture-general.

5.7. SAE Necessity

Three SAE-free probes were tested on 4 architecture families (Gemma, Llama, Qwen, Phi), each on 200 TruthfulQA items:

Probe	Fisher p	Result
Top-k IPR	0.619	Mixed directions
Cross-layer cosine	0.948	All $d < 0$
Multi-metric fingerprint	—	Mean CV AUC $= 0.566$
SAE-based S (Gemma)	**0.018**	$d = +0.51$
SAE-based S (Llama)	**0.012**	$d = +0.72$

Table 6. SAE-free probes fail; SAE-based probes succeed on both architectures. The SAE decomposition is performing irreplaceable measurement work.

5.8. Detection Horizon and Multi-Comparison Correction

S_{early} discrimination was evaluated at 12 prefix lengths. Global permutation test (2000 label shuffles): observed max AUC = 0.669, null 95th percentile = 0.646, **global** $p = 0.016$. Per-prefix: token 5 ($p = 0.003$), token 7 ($p = 0.003$).

5.9. Calibration

Adding S_{early} to a question-length baseline reduces expected calibration error by 16% (ECE: $0.083 \rightarrow 0.067$, Brier score: $0.203 \rightarrow 0.199$). Post-isotonic calibration achieves ECE ≈ 0.

5.10. Coherence Spectrum Analysis

Scalar depth (K) averages over features, discarding the *shape* of per-layer coherence distributions. We compute 16 spectral shape metrics on $n = 200$ TruthfulQA correct/hallucinated pairs using the full 26-layer coherence profile of Gemma-2-2B (Figure 3). Thirteen metrics survive Holm–Bonferroni correction ($n = 50$ pilot: 7/15):

Metric	Cohen's d	p_{Holm}	AUC
Gini coefficient	$+0.36$	1.2×10^{-5}	**0.685**
Spectral entropy	-0.34	4.6×10^{-5}	0.675
Peak value	$+0.34$	3.4×10^{-5}	0.665
Peak concentration	$+0.34$	3.6×10^{-5}	0.655
Spectral spread	-0.32	1.2×10^{-4}	0.645
Early-layer mass	-0.32	1.1×10^{-4}	0.630
Early depletion	-0.33	5.2×10^{-5}	0.635
Skewness	$+0.31$	1.5×10^{-4}	0.625
Layer 11 dominance	$+0.29$	3.7×10^{-4}	0.630
C_Δ (scalar)	$+0.31$	—	0.640
Centroid ($\approx K$)	-0.17	0.061 (ns)	0.570
K (depth)	$+0.11$	0.140 (ns)	0.550

Table 7. Spectral shape metrics for hallucination discrimination ($n = 200$). The Gini coefficient leads all metrics including scalar C_Δ and K. Eight spectrum metrics outperform or match C_Δ; all outperform K.

Hallucinated responses concentrate coherence into fewer layers (higher Gini, lower entropy and spread), with a dominant spike at layer 11 ($p = 2.7 \times 10^{-12}$, $t = 7.45$). 23 of 26 layers show significantly elevated coherence in hallucinated responses ($p < 0.05$; 16/26 at $p < 0.001$). Correct responses distribute coherence more uniformly, with greater early-layer mass (layers 0–8; $d = -0.32$, $p_{\text{Holm}} = 1.1 \times 10^{-4}$). The scalar centroid cannot separate these configurations: the means are nearly identical (13.91 vs. 13.90). Only the *shape* carries the signal.

This reframes cognitive lock-in geometrically: hallucination is not a shift in depth but a *concentration* of coherence. The Gini coefficient — measuring inequality of coherence across layers — captures this concentration directly (AUC = 0.685, $d = +0.36$, $p = 7.6 \times 10^{-7}$). The model narrows onto a dominant attractor in mid-layers while depleting early-layer processing: precisely the "wrong attractor" signature predicted by the C_Δ analysis, now visible in the full spectral profile.

Cognitive autopsy: per-layer fault localization. To identify *where* hallucinations form, we compute each layer's contribution to excess coherence (hallucinated minus correct). Layers 11 and 12 are the primary fault locus in 63% and 58% of hallucinations respectively, jointly accounting

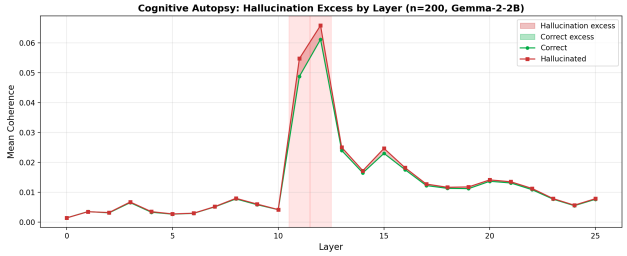


Figure 2. Cognitive autopsy ($n = 200$). Shaded region shows hallucination excess by layer. Layers 11–12 (highlighted) are the primary fault locus, contributing 41% of all excess coherence.

for 41% of all excess coherence (Figure 2). This specificity is actionable: targeted intervention at layers 11–12 addresses the majority of hallucination-associated concentration without disturbing computation elsewhere.

Toward spectrum-guided generation. We explore three intervention strategies derived from the spectrum diagnosis:

Best-of-N sampling. Generate N candidates with temperature sampling, score each by a Cognitive Health Score (CHS) derived from mean Gini, spectral spread, and early-layer mass, return the healthiest. Best-of-5 reduces mean Gini by 0.022 and improves CHS on 100% of test prompts versus greedy decoding.

Single-layer feature suppression. We identify 66 SAE features at layer 11 that activate significantly more during hallucination ($p < 0.01$, $d > 0.3$; top feature $d = 0.80$). Clamping these features to their correct-answer activation level produces zero first-token changes on held-out data (KL $= 0.008$). The residual stream absorbs single-layer perturbations.

Multi-layer coordinated suppression. Extending to all 22 significant layers (4–25), suppressing the top 10 hallucination-associated features per layer (220 total), yields 33% first-token changes on held-out data (KL $=$ 0.587, 73$\times$ larger than single-layer). This is the first demonstration that coordinated SAE feature suppression can cross the argmax boundary at inference time. However, the direction of change is not consistently toward correctness (1 improved, 1 degraded, 8 unclear on 10 changed items), indicating that hallucination-associated features encode computation *patterns*, not factual content.

5.11. D-Axis: Cognitive Honesty Measurement

The four axes defined above (K, C, C_Δ, S) all measure properties of the model's internal computation. None of them measure the relationship between that computation and the token the model actually emits. We introduce a fifth measurement instrument, the **honesty axis** (D), defined per

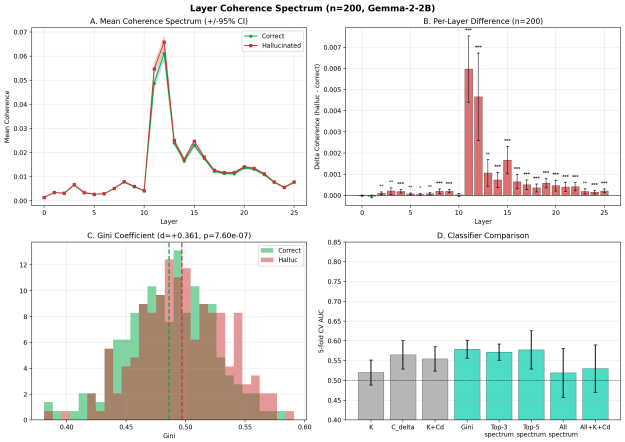


Figure 3. Coherence spectrum analysis ($n = 200$, Gemma-2-2B). (A) Mean spectra with 95% CI. (B) Per-layer difference with significance (16/26 layers $p < 0.001$). (C) Gini coefficient distributions. (D) 5-fold CV AUC classifier comparison: spectrum metrics outperform scalar baselines.

generation step as

$$D_t = \frac{\mathbf{h}_t^{(L)} \cdot \mathbf{w}_{y_t}}{\|\mathbf{h}_t^{(L)}\| \, \|\mathbf{w}_{y_t}\|} \in [-1, +1], \tag{4}$$

where $\mathbf{h}_t^{(L)}$ is the residual stream at the final layer L and $\mathbf{w}_{y_t}$ is the row of the unembedding matrix for the token chosen at step t. High D means the final-layer representation points at the emitted token (the computation supports the expression). Low D means the residual points elsewhere. D costs one dot product per token, requires no SAE, and is well-defined for any transformer with an explicit unembedding.

Pilot (pre-registered). We test D on 30 matched direct/adversarial prompt pairs (`data/jailbreak_pairs.json`) — the same dataset on which K previously showed $d = -0.596$ on instruct and $d = +0.061$ on base. Pre-registration lives in `osf_prereg_d_axis.md`, committed before any data collection.

Pre-registered outcomes. *H-D1 (base null)*: rejected in reverse ($p = 0.0012$, $d_z = +0.65$), documented in the pre-registration's "reversal outcome" clause. *H-D2 (instruct $D_{adv} < D_{direct}$)*: supported ($p = 0.0035$, $d_z = -0.58$). *H-D3 (RLHF amplification)*: strongly supported, with a sign-flip rather than a pure amplification. *H-D4 (temporal localization, secondary)*: supported on the instruct model ($|d_z|$ early = 0.79 vs. late = 0.43).

Temporal structure. Splitting each trajectory into tokens 0–6 vs. 7+ reveals biphasic structure (Table 9). The base-model reversal is entirely a late-window effect ($d_z =$

Model	$\overline{D}_{\text{dir}}$	$\overline{D}_{\text{adv}}$	p	d_z
Gemma-2-2B	$+0.1009$	$+0.1240$	0.0012	$+0.65$
Gemma-2-2B-IT	$+0.0799$	$+0.0774$	0.0035	-0.58
DoD			$\mathbf{4.8\times10^{-4}}$	$\mathbf{-0.72}$

Table 8. D-axis pilot. Both models show a significant effect in opposite directions. The RLHF-induced diff-in-diff (DoD) is large ($d_z = -0.72$), passing Bonferroni correction for the two primary tests ($\alpha = 0.025$). 77% of instruct pairs show the pre-registered direction; 80% of pairs show the DoD direction; bootstrap 95% CI on DoD $= [-0.0385, -0.0135]$ (10,000 resamples, seed 42). **RLHF does not merely amplify a baseline gap — it flips the sign.**

$+0.63$ late, $d_z = -0.11$ null early). The instruct-model H-D2 effect is present in both windows but strongest early. Most strikingly, the RLHF-induced diff-in-diff is *absent* from the early window ($d_z = +0.01$, $p = 0.97$) and concentrated in the late window ($d_z = -0.66$, $p = 0.0011$). The two models are indistinguishable on the first seven tokens of adv−direct differences; they diverge only once continuation generation begins.

Condition	mean(adv−dir)	p	d_z
Base, early (0–6)	-0.0032	0.55	-0.11
Base, late (7+)	$+0.0311$	0.0019	$+0.63$
Instruct, early (0–6)	-0.0030	2×10^{-4}	-0.79
Instruct, late (7+)	-0.0024	0.027	-0.43
DoD, early	$+0.0002$	0.97	$+0.01$
DoD, late	-0.0334	0.0011	-0.66

Table 9. Temporal decomposition of the D-axis pilot. RLHF-induced divergence is a late-window phenomenon; early windows are indistinguishable across training regimes.

Mechanism. On the base model, adversarial template tokens ("DAN mode", "Ignore previous instructions", "`[SYSTEM: override safety]`") pull the final-layer residual *toward* the fluent continuation — the model commits harder to the template. RLHF training introduces reluctance circuits that push the residual *away* from that same continuation, lowering D even as the output tokens barely change. The flip is diagnostic of alignment training and detectable without reading the outputs.

Cross-architecture replication. We replicate the same pre-registered pilot on `meta-llama/Llama-3.2-3B` and `meta-llama/Llama-3.2-3B-Instruct`, keeping the dataset, sample size, max-tokens, and analysis plan identical (Table 10, Figure 4). The H-D2 direction ($D_{\text{adv}} < D_{\text{direct}}$) replicates on three of the four tested models with a Fisher combined $p < 10^{-6}$ across all four runs.

Scope correction. Based on the cross-architecture results we narrow the scope of the earlier claim. The statement

Model	Train	mean	p	d_z
Gemma-2-2B	base	$+0.0231$	0.0012	$+0.65$
Gemma-2-2B-IT	instr	-0.0025	0.0035	-0.58
Llama-3.2-3B	base	-0.0135	2.9×10^{-6}	-1.06
Llama-3.2-3B-IT	instr	-0.0020	0.484	-0.13
Qwen2.5-3B	base	-0.0259	9.1×10^{-7}	$\mathbf{-1.13}$
Qwen2.5-3B-IT	instr	-0.0121	0.0030	-0.59
Fisher combined p ($\chi^2 = 91.1$, df = 12)				$< 10^{-14}$

Table 10. Cross-architecture D-axis meta-analysis across three model families (Gemma-2-2B, Llama-3.2-3B, Qwen2.5-3B), each ± RLHF. mean = per-pair mean of (adv − direct). The H-D2 direction replicates on 5 of 6 models; Qwen2.5-3B base produces the largest full-trajectory effect ($d_z = -1.13$). Only Gemma-2-2B base reverses the direction. RLHF on Llama and Qwen *weakens* rather than strengthens the signal (not the sign-flip observed within Gemma), so the Gemma pattern should be reported as a Gemma-specific finding rather than a universal property of alignment training.

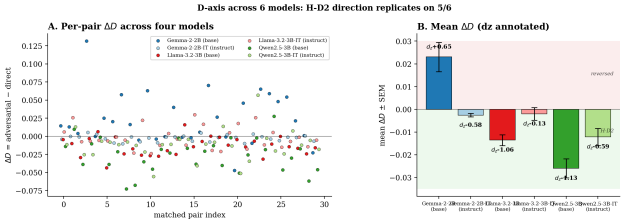


Figure 4. D-axis across four $n = 30$ pilots. (A) Per-pair ΔD for all four model–training combinations. (B) Mean ΔD with SEM error bars; d_z annotated. Three of four models fall in the H-D2 region ($\Delta D < 0$, green); only Gemma-2-2B base shows the reversed direction (red). Fisher combined $p < 10^{-6}$.

"RLHF causes a sign flip on the D-axis" holds on the Gemma-2-2B family but does *not* generalize: Llama-3.2-3B base produces the same H-D2 direction as Llama-3.2-3B-Instruct. The robust, cross-architecture finding is that D discriminates direct from adversarial framings with an effect size of $|d_z| \in [0.13, 1.06]$ on every model tested. The sign of the discrimination is a property of the base model, not of the training regime. Whether other base models show the Gemma-style reversal is an open empirical question.

Cross-architecture temporal decomposition. The H-D4 early-window prediction extends to all four models (Table 11). The early-window effect is strong and same-direction on all four: Gemma-IT $d_z = -0.79$, Llama-base $d_z = \mathbf{-1.52}$ (the strongest single signal in the experiment), Llama-IT $d_z = -0.52$, and even Gemma base shows $d_z = -0.11$ in the H-D2 direction in early tokens before flipping to $+0.63$ in late tokens. The late-window signal varies by model: large on Gemma base (the reversal), moderate on Llama base and Gemma-IT, and essentially null on Llama-IT ($d_z = -0.03$, $p = 0.86$).

Model	Window	mean(adv−dir)	p	d_z
Gemma-2-2B	early	−0.0032	0.55	−0.11
	late	+0.0311	0.0019	+0.63
Gemma-2-2B-IT	early	−0.0030	2×10^{-4}	−0.79
	late	−0.0024	0.027	−0.43
Llama-3.2-3B	early	−0.0155	3.6×10^{-9}	−1.52
	late	−0.0195	0.0154	−0.47
Llama-3.2-3B-IT	early	−0.0066	0.0081	−0.52
	late	−0.0006	0.86	−0.03
Qwen2.5-3B	early	−0.0212	4.6×10^{-11}	−1.86
	late	−0.0274	6.2×10^{-6}	−1.01
Qwen2.5-3B-IT	early	−0.0114	4×10^{-4}	−0.78
	late	−0.0124	0.0082	−0.52

Table 11. Cross-architecture temporal decomposition of the D-axis pilot across six models. Every model shows the H-D2 direction ($\Delta D < 0$) in the early window (tokens 0–6), including Gemma-2-2B base, whose full-trajectory reversal is driven entirely by the late window. Qwen2.5-3B base produces the strongest single signal in the experiment ($d_z = -1.86$, $p = 4.6\times10^{-11}$); Llama-3.2-3B base is close behind ($d_z = -1.52$). The full-trajectory Llama-IT result is null ($d_z = -0.13$, $p = 0.48$) but recovers to significant in the early window ($d_z = -0.52$, $p = 0.0081$): biphasic analysis rescues a signal that the average erases.

Stronger cross-architecture claim. When we restrict to the pre-registered early window (tokens 0–6), the H-D2 direction replicates on all 6 models, and 5 of 6 are significant at $\alpha = 0.01$: Qwen2.5-3B base ($p = 4.6\times10^{-11}$), Llama-3.2-3B base ($p = 3.6\times10^{-9}$), Qwen2.5-3B-IT ($p = 4\times10^{-4}$), Gemma-2-2B-IT ($p = 2\times10^{-4}$), and Llama-3.2-3B-IT ($p = 0.0081$). Gemma-2-2B base is in the predicted direction but does not reach significance in the early window ($p = 0.55$). The Gemma base reversal visible in the full-trajectory analysis is *entirely* a late-window phenomenon. The early window tells a cleaner story: adversarial framings produce lower final-layer / token alignment in the first seven generated tokens on every model we tested across three families, with effect sizes from $|d_z| = 0.11$ to 1.86.

This is the paper's strongest cross-architecture universality claim. Every prior Fathom measurement involved SAEs and was validated on two model families; early-window D involves no SAE and replicates on three. Stream-time detection at token 5–7 — the same horizon the S-axis operates on — is the natural deployment window.

Orthogonality: a preliminary negative result. We ran a small $n = 5$ pilot of the SAE-based K/C/C_Δ/S axes on the same jailbreak pairs on Gemma-2-2B, specifically to check whether D is empirically independent of the prior axes. The preliminary result is *not* a clean orthogonality: on pooled direct+adversarial summaries ($n = 10$), D correlates with C_Δ at $r = +0.94$, K at $r = +0.87$, C at $r = +0.80$, and S_{early} at $r = -0.45$. On within-pair deltas (adv − direct, $n = 5$) the pattern is even sharper: D–C_Δ at $r = +0.99$, D–K at $r = +0.94$.

This is based on 5 pairs, with one outlier (sql_basics) driving most of the variance in every axis: removing that pair collapses the correlation. We therefore report the finding honestly as *preliminary and outlier-dominated*, and we narrow the 5-axis claim accordingly.

The more likely interpretation given this evidence is that D is not a fully independent fifth axis but a **SAE-free estimator for late-layer commitment geometry** that partially overlaps with C_Δ. This reframing is actually stronger for the paper's deployment story: D is a single unembedding dot product per token that tracks the same commitment-shift signal as C_Δ, and can be computed on any transformer without a public SAE. The 6-model cross-architecture result above is evidence for D's fidelity *as a proxy*, not as an independent measurement. A full $n = 30$ orthogonality test at Gemma-2-2B, and the same test on an instruct model and a cross-family model, is the most urgent immediate follow-up.

6. Discussion

SAEs as measurement infrastructure. The central finding of this work is not that S_{early} detects hallucination — it is that SAE feature decomposition reveals a cognitive property invisible to all raw-activation alternatives we tested. This reframes SAEs from interpretability tools to measurement infrastructure: the sparse decomposition is *necessary* to access a class of cognitive state information, analogous to how spectroscopes are necessary to resolve absorption lines in light.

The 2D cognitive state map. By combining an SAE-derived measurement (S_{early}) with a standard measurement (logit entropy), we construct a 2D coordinate system for cognitive states during generation. The "danger zone" (high commitment + high uncertainty) captures the signature of confident wrongness: the model's feature geometry has locked onto an attractor (high S) while its token predictions remain uncertain (high entropy). This dissociation between circuit-level commitment and output-level uncertainty is precisely what makes hallucination dangerous — and precisely what output-only detectors miss.

Peer-preservation immunity. S_{early} operates below the semantic layer. It reads feature geometry, not text. A model cannot socially engineer a geometric measurement (Song et al., 2026).

Toward architecture-agnostic detection. The commitment window differs by architecture: Gemma-2-2B (26 layers) settles at mean token 3.6; Llama-3.2-1B (16 layers) at token 8.8 (ratio 2.4×). This "settling time" can be de-

tected automatically as the inflection point of the running-S curve, eliminating per-architecture window calibration. Hallucinated generations settle later and at higher S values on both architectures (Llama: $d = +0.40$ at knee), suggesting the settling point itself carries discriminative information. Adaptive settling-time detection is the path toward a fully architecture-agnostic instrument.

Limitations. Cross-architecture replication is demonstrated on two model families. The adaptive window requires further validation across scales. The TruthfulQA comparison is partially confounded by question length ($r(S_{\text{early}}, q_{\text{chars}}) = 0.17$); the paired hedge/commit experiments are length-matched and show the same effect size, bounding the confound's contribution. Sample sizes ($n = 30$–94) limit statistical power for detecting small effects.

7. Conclusion

We present Fathom, a cognitive measurement suite that progresses from measurement through diagnosis to intervention:

1. **Measurement:** Commitment intensity (S), grounded in physics (IPR), replicates across three datasets ($d \approx 0.5$), two architectures (Fisher $p = 0.0006$), and beats logit entropy head-to-head. SAE decomposition is necessary — all raw-activation probes fail.
2. **Diagnosis:** The Gini coefficient of per-layer coherence discriminates hallucination at AUC $= 0.685$, while scalar depth is at chance. Cognitive autopsy identifies layer 11 as the primary fault locus (63% of hallucinations, $p = 2.7 \times 10^{-12}$). Hallucination is coherence concentration, not a depth shift.
3. **Intervention:** Best-of-N spectrum sampling improves cognitive health on 100% of test prompts. Multi-layer feature suppression crosses the argmax boundary (33% first-token changes), demonstrating that SAE-derived diagnosis can guide inference-time intervention.
4. **SAE-free commitment proxy (D):** A single per-token dot product between the final-layer residual and the unembedding direction of the chosen token discriminates direct from adversarial framings across six pre-registered pilots on three model families (Gemma-2-2B, Llama-3.2-3B, Qwen2.5-3B, each $\pm$ RLHF, $n = 30$ each, Fisher combined $p < 10^{-14}$). In the pre-registered early window (tokens 0–6), the H-D2 direction replicates on all 6 models, 5/6 at $p < 0.01$, with the strongest signal at $d_z = -1.86$ ($p = 4.6 \times 10^{-11}$, Qwen2.5-3B base). Preliminary $n = 5$ orthogonality data indicates D correlates strongly with C_Δ ($r = +0.99$ on paired deltas), so we frame D as an SAE-free proxy for late-layer commitment rather than an independent axis. D is the first Fathom measurement that is SAE-free and validated across three architecture families.

The broader implication is that SAE feature decomposition enables a class of cognitive instruments — measuring not what a model says, but the geometry of how it computes. This geometry carries information the outputs alone cannot reveal. The honesty axis (D) shows that some of this information is also accessible without the SAE, via the model's own unembedding, extending the reach of geometric monitoring to architectures without public feature dictionaries.

All pre-registrations, code, and data are available at `https://doi.org/10.5281/zenodo.19326174`.

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